

Fullstack Assignment: Image Tagger

Overview

Your task is to build a fullstack app. Users must be able to sign up, log in, generate random images, and place "pins" on them to create comment threads.

Backend API Requirements

- **Data:** Use in-memory storage (e.g., arrays).
- **IDs:** Use **ULIDs** for all new resources.
- **Auth:** Implement simple, stateless authentication. Protected routes require a valid username passed in a custom **X-User-Name** header. No JWT or sessions needed.
- **Security:** Sanitize all user-submitted text (comments) on the backend.

Endpoints

User

- **POST /users** - Create user. Body: `{"name": "<string>"}`.
- **POST /login** - Authenticate user. Body: `{"name": "<string>"}`. Returns 200 if user exists, 401 otherwise.

Protected Endpoints (require **X-User-Name** header)

- **GET /users** - Retrieve all users.
- **POST /images** - Create new image. Returns `{id: <ULID>, url: "https://picsum.photos/id/..."}`.
- **GET /images** - Retrieve list of all images.
- **POST /images/<:id>/threads** - Create a new comment pin. Body: `{"x": <number>, "y": <number>, "comment": "<string>"}`.
- **GET /images/<:id>/threads** - Retrieve all threads for a specific image.
- **DELETE /threads/<:id>** - Delete a thread (must be creator).



Frontend UI Requirements

- Use any modern framework or vanilla JS.

Pages & Auth

- `/signup` page: Username field. On success, log in and redirect to `/`.
- `/login` page: Username field. On success, store username (for auth headers) and redirect to `/`.
- Main app (`/`) must be protected from unauthenticated users.

Main App Layout

- **Header:**
 - "Generate New Image" button.
 - "Comment Mode" toggle switch.
 - User display (`Welcome, [Name]!`) and "Sign Out" button.
- **Main Content (2-column):**
 - **Left Sidebar:** Linkable list of all generated image IDs.
 - **Content Viewer:** Displays the selected image.

Core Functionality

- Clicking "Generate New Image" creates an image and auto-navigates to it.
- Clicking an image ID in the sidebar loads it in the viewer.
- Image must scale to fit its container and be responsive to window resizing.
- **Comment Mode (On):** Clicking the image opens a text dialog. Submitting creates a new pin at that (x, y) coordinate.
- **Comment Mode (Off):** Clicking the image does nothing.
- Pins on the image must display the first 2 letters of the creator's name.

★ Bonus Features

1. **Image Deletion:** Implement `DELETE /images/<:id>` and add a delete button in the UI.
2. **Draggable Pins:** Allow users to drag existing pins. Update the pin's (x, y) coordinates on the backend when dropped.

Submission

- Provide a link to a single Git repository.
- Include a `README.md` with clear setup and run instructions for both frontend and backend.